

Jenkins EC2 + Docker Hub + EKS CI/CD Demonstration Guide

Architecture:

```
Developer
|
v
GitHub Repository (pipeline1)
|
v
Jenkins Server (EC2)
|
+--> Git Checkout
+--> Docker Build
+--> Docker Push
|
v
Docker Hub
|
+--> kubectl deployment
|
v
Amazon EKS Cluster (eksdemo1)
|
v
Kubernetes Pods
```

Step 1: Prepare Jenkins EC2

Install Docker:

```
sudo apt update
sudo apt install docker.io -y
```

Start Docker:

```
sudo systemctl start docker
sudo systemctl enable docker
```

Allow Jenkins user:

```
sudo usermod -aG docker jenkins
```

Restart Jenkins:

```
sudo systemctl restart jenkins
```

Step 2: Install kubectl

```
curl -LO https://dl.k8s.io/release/v1.30.0/bin/linux/amd64/kubectl
chmod +x kubectl
sudo mv kubectl /usr/local/bin/
kubectl version --client
```

Step 3: Configure AWS Access for EKS

Configure AWS CLI:

```
aws configure
```

Update kubeconfig:

```
aws eks update-kubeconfig --name eksdemo1 --region ap-south-1
```

Verify:

```
kubectl get nodes
```

Step 4: Clone GitHub Repository

Repository:

```
https://github.com/raushan8586/pipeline1.git
```

Commands:

```
git clone https://github.com/raushan8586/pipeline1.git
```

```
cd pipeline1
```

Step 5: Docker Build and Push to Docker Hub

Build:

```
docker build -t pipeline1:v1 .
```

Login:

```
docker login
```

Tag:

```
docker tag pipeline1:v1 yourdockerhubuser/pipeline1:v1
```

Push:

```
docker push yourdockerhubuser/pipeline1:v1
```

Step 6: Kubernetes Deployment

deployment.yaml should use:

```
image:
  yourdockerhubuser/pipeline1:v1
```

Apply:

```
kubectl apply -f k8s -n demo
```

Verify:

```
kubectl get pods -n demo
kubectl get svc -n demo
```

Step 7: Jenkins Docker Hub Credentials

Jenkins:

```
Manage Jenkins
-> Credentials
-> Add Credentials
```

Kind:

Username with password

ID:

dockerhub-creds

Use Docker Hub username and access token.

Step 8: Jenkinsfile Pipeline

Pipeline stages:

1. Checkout code from GitHub
2. Build Docker image
3. Login to Docker Hub
4. Push image
5. Deploy to EKS

Example deployment command:

```
kubectl set image deployment/pipeline1 pipeline1=image:tag -n demo
```

```
kubectl rollout status deployment/pipeline1 -n demo
```

Final CI/CD Flow

```
Git Push
|
v
Jenkins EC2
|
+--> Build Docker Image
|
+--> Push Docker Hub
|
+--> kubectl update
|
v
EKS Rolling Deployment
```

This demonstrates:

- GitHub
- Jenkins CI/CD
- Docker
- Docker Hub
- Kubernetes
- Amazon EKS